AI & IOT INTEGRATED WIND TREE FOR OPTIMIZED MICRO WIND ENERGY

Presented by:

M. Thanmai(22491A05G6)
G. Sri Sowmya(22491A05E0)
G. Aksha(22491A0583)
K. Zubair(22491A05E8)
K. Naga Jyothi(22491A05F8)
K. Moulya Sri(22491A05E9)

Under the Guidance of

D.Vidyanandha Babu
Professor
Department of ------------------------------

Outline

- ABSTRACT
- INTRODUCTION
- LITERATURE REVIEW
- OBJECTIVE
- EXISTING METHOD
- PROPOSED METHOD
- HARDWARE & SOFTWARE INVOLVED
- ADVANTAGES AND APPLICATION
- REFERENCES

ABSTRACT

- This project explores an innovative, nature-inspired approach to renewable energy generation by developing a wind tree system utilizing aero leaves.

- Designed to mimic the structure and aesthetics of a natural tree, this setup integrates small, lightweight aero-leaf turbines in place of leaves to capture wind from all directions.

- With the integration of “Internet of Things technology and AI-based monitoring”, the system ensures efficient energy tracking, fault detection, and real-time performance optimization.

- This model emphasizes affordability and accessibility, promoting clean energy adoption at the grassroots level.

INTRODUCTION

- Wind Tree technology is a nature-inspired solution that mimics the form and function of natural trees to harness wind energy efficiently using compact vertical axis turbines called aeroleaves.
- Each aeroleaf acts as a miniature wind turbine that can generate electricity even from low-speed winds in all directions, making it suitable for areas where traditional turbines are inefficient.
- Integrating Artificial Intelligence (AI) and Internet of Things (IoT) enhances real-time monitoring, predictive maintenance, and adaptive energy optimization in the system.
- This Project explores how combining smart systems and renewable energy principles can lead to a scalable and eco-friendly power generation solution.

LITERATURE SURVEY

Ref. No	Title & Author	Technique Used	Pros	Cons
1	Fabrication and performance analysis of the aero-leaf savonius wind turbine tree	Savonius rotor design, performance testing	Works efficiently, urban-friendly	Mechanical imbalance, require external battery
2	Nature-inspired designs in wind energy	Bio-mimicry analysis, computational simulation	Low-noise, useful for urban architecture	May increase fabrication complexity and cost
3	Small wind turbines and their potential for internet of things applications	Design & Evaluation of Small Wind Turbines	Sustainable Power Source for IOT	Wind Availability Constraints Low Power Output
4	Energy prediction of wind turbine using iot	Sensor Integration with IoT	Real-time Monitoring, Energy Forecasting	Limited Sensor Accuracy, Model Limitations
5	Optimizing Energy Flow in a Wind-PV-Battery Microgrid Using AI Techniques	Microgrid, Energy Management, Swarm Intelligence, Wind Power, Photovoltaics	Optimized energy flow means reduced dependency on expensive fossil-fuel backups and lower operational costs	Installing wind turbines, solar panels, battery systems, and sensors, along with AI infrastructure, requires a significant upfront investment.

Ref. No	Title & Author	Technique Used	Pros	Cons
6	A comprehensive IoT cloud-based wind station ready for real-time measurements and artificial intelligence integration	Weather station Wind measurements IoT wind station Cloud-base weather monitoring	Immediate awareness of wind conditions and turbine performance	Requires investment in IoT hardware, cloud services, and AI development.
7	Review of Artificial Intelligence-Based Design Optimization of Wind Power Systems	Evolutionary Algorithms (GA, PSO), Hybrid AI Approaches	-Comprehensive coverage of optimization areas	- Limited discussion on real-time deployment
8	AI-Enabled Metaheuristic Optimization for Predictive Management of Renewable Energy Production in Smart Grids	Hybrid Metaheuristics (LSTM-RL, RL-SA, CNN-PSO)	- Suitable for load balancing and rural microgrids	- May require significant training data
9	Internet of Things (IOT) Based Machine Learning Techniques for Wind Energy Harvesting	IoT Sensors, Wireless Sensor Networks (WSNs)	Predictive Maintenance, Energy Output Optimization	High Initial Cost, Data Security Risks
10	Small Wind Electric System Energy Saver	Permanent Magnet Generators (PMG)	Energy Independence, Easy Installation	Noise & Vibration, Limited Output

MOTIVATION

- Most wind turbines require large open areas and consistent wind flow, making them unsuitable for cities, rooftops, or rural households with limited land.

- Inspired by natural leaf structures, the concept of a "Wind Tree" using aero-leaves offers a new path that is both functional and aesthetic.

- There is limited practical implementation data on nature-inspired vertical axis turbines like the aero-leaf design — making this an opportunity to contribute real-world testing and improvement.

- The project aligns with global efforts to promote green, decentralized, and noise-free energy solutions that are safe for humans, animals, and the environment.

OBJECTIVE

- The Main motive of this project is to design and fabricate a low-cost Wind Tree prototype that harnesses wind energy using multiple aero-leaves arranged in a tree-like structure.
- To integrate IoT sensors for real-time monitoring of wind speed, turbine RPM, and energy generation statistics.
- To apply basic AI or Machine Learning models to predict energy output patterns and optimize blade performance under varying wind conditions.

EXISTING METHOD

- Earlier introduced project include a wind turbine model shaped like a tree, using aero-leaf
  shaped Savonius rotors as the main energy-generating components.
- Each aero-leaf works as a small vertical-axis wind turbine (VAWT), capturing wind from all directions.
- The design focuses on low wind speed environments, making it ideal for urban and semi-urban use.
- The rotors are connected to small DC generators to produce electricity.

EXISTING METHOD

- The existing method compares traditional turbine designs with bio-inspired models, highlighting how natural shapes improve airflow and reduce noise.

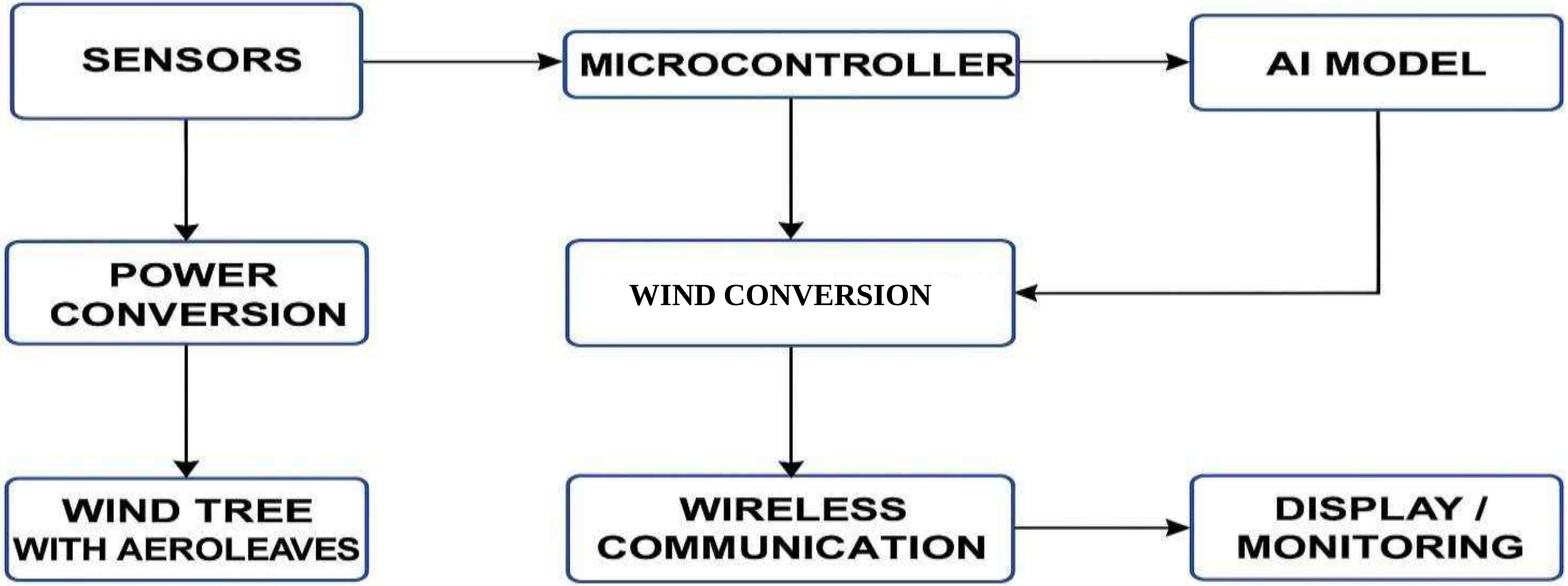
- The focus is on biomimicry i.e., copying nature’s design principles to improve wind energy generation.

- It contains aesthetic integration, allowing turbines to blend with parks, schools, or cityscapes without disturbing surroundings.

PROPOSED METHOD

- Design and build a small-scale Wind Tree using aero-leaf-shaped Savonius rotors.
- Using recycled plastic, DC motors to keep minimal or zero cost.
- Integrating IoT sensors to monitor wind speed, power output, and rotor performance in real-time.
- Develop a machine learning model to predict energy generation based on historical weather and turbine data.

BLOCK DIAGRAM



EXPLANATION OF PROPOSED WORK

What is the use?

- This project focuses on designing a nature-inspired wind energy harvesting system called a Wind Tree, helps reduce dependence on fossil fuels by using naturally available wind.

- Unlike traditional wind turbines, this system is compact, silent, and suitable for urban and rural areas where space and aesthetics are concerns.

- The system integrates Artificial Intelligence (AI) for efficient wind prediction and system optimization, while IoT (Internet of Things) is used for real-time monitoring of power generation, turbine health, and environmental data.

How it Going to be used in your Project?

- All the aero-leaves are connected to small DC generators. These will generate power that can be stored in batteries for later use.
- Connecting sensors to the system to monitor wind speed, turbine rotation, and energy generation. This data will be sent to a mobile or web dashboard for live tracking.
- Using machine learning models, we will predict the best time and conditions for maximum energy generation based on past data.
- It is a working model of the wind tree will be developed using recycled or low-cost materials to demonstrate the process from wind capture to energy output.

Why You Choose and explain the advantages ?

The reason behind choosing this project is,

- With rising pollution and climate change, there’s a strong need to shift from fossil fuels to eco-friendly energy solutions like wind power.

- Unlike big wind turbines, this project works even with low or irregular wind flow, which is ideal for cities and villages.

- The idea of using a "wind tree" with aero-leaves is unique, visually appealing, and blends well with the environment.
- This project can be built using low-cost, reused items, making it affordable for students and communities.

HARDWARE & SOFTWARE INVOLVED

HARDWARE:

- Aero-Leaves (Mini Savonius Turbines)
- DC Micro Generators
- Battery
- Microcontroller
- Charge Controller
- Sensors: Anemometer – Measures wind speed RPM Sensor – Tracks turbine rotation speed. Voltage & Current Sensors – Monitor power generation.
- Wi-Fi/Bluetooth Module
- Mounting Tree Structure

SOFTWARE

- Arduino IDE

For programming microcontrollers and reading sensor data.

- ThingSpeak

Cloud-based platform for real-time IoT data visualization.

- Proteus

For circuit simulation before real hardware testing.

- Python / MATLAB

Used for data analysis, predictions, and AI-based forecasting.

- MicroPython

For coding the microcontroller logic.

APPLICATION & ADVANTAGES

APPLICATION:

- This project supports sustainable infrastructure by powering streetlights, CCTV cameras, and IoT devices.
- It can provide basic electricity in remote villages or forest surveillance posts.
- It can acts as a natural energy backup for LED bulbs, fans, mobile charging, etc.
- It can be used in Environmental Monitoring System.

ADVANTAGES:

- It works silently unlike traditional horizontal turbines.
- We can increase the number of leaves (turbines) easily as per energy needs.
- Real-time power prediction, fault detection, and performance tracking from your phone or laptop.
- It blends well in public places without disturbing the visual appeal.

REFERENCES

1. Patidar, H.; Shende, V.; Baredar, P.; Soni, A. Comparative study of offshore wind energy potential assessment using different Weibull parameters estimation methods. *Environ. Sci. Pollut. Res.* 2022, *29*, 46341–46356. [CrossRef]
2. Bp Statistical Review of World Energy. 2022. Available online: https://www.bp.com/en/global/corporate/energy-economics/ statistical-review-of-worldenergy.html (accessed on 15 January 2023).
3. Power Consumption to Triple by 2050*0* —McKinsey Report. Available online: https://www.modernpowersystems.com/news/ newspower-consumption-to-triple-by-2050-mckinsey-report-9655086# (accessed on 8 January 2023).
4. Amano, R.S. Review of Wind Turbine Research in 21st Century. *J. Energy Resour. Technol.* 2017, *139*, 050801. [CrossRef]
5. Prabhukhot, P.R.; Prabhukhot, A.R. Computer Analysis of S822 Aerofoil Section for Blades of Small Wind Turbines at Low Wind Speed. *J. Sol. Energy Eng.* 2017, *139*, 051008–51011. [CrossRef]
6. Global Energy Review 2020 by IEA. Available online: https://www.iea.org/reports/global-energy-review-2020 (accessed on 15 January 2023).
7. using numerical and metaheuristic optimization methods for different Indian terrains. *Environ. Sci. Pollut. Res.* 2023, *30*, 30874–30891. [CrossRef]

REFERENCES

8. Mishnaevsky, L.; Hasager, C.B.; Bak, C.; Tilg, A.-M.; Bech, J.I.; Rad, S.D.; Fæster, S. Leading edge erosion of wind turbine blades: Understanding, prevention and protection. *Renew. Energy* 2021, *169*, 953–969. [CrossRef].
9. Chandrakant, S.; Panchal, H.; Sadasivuni, K.K. Numerical simulation of flow-through heat exchanger having helical flow passage using high order accurate solution dependent weighted least square based gradient calculations. *Energy Sources Part A Recover. Util. Environ. Eff.* 2021, 1–26. [CrossRef]
10. Tian, W.; Song, B.; VanZwieten, J.H.; Pyakurel, P. Computational Fluid Dynamics Prediction of a Modified Savonius Wind Turbine with Novel Blade Shapes. *Energies* 2015, *8*, 7915–7929. [CrossRef]
11. Patel, A.; Soni, A.; Baredar, P.; Malviya, R. Analysis of temperature distribution over pipe surfaces of air-based cavity linear receiver for cross-linear concentration solar power system. *Environ. Sci. Pollut. Res.* 2022, 1–19. [CrossRef] [PubMed]
12. Blackwell, B.; Sheldahl, R.; Feltz, L. *Wind Tunnel Performance Data for Two-and-Three-Bucket Savonius Rotors*; US Sandia Laboratories Report SAND76-0131; Sandia Laboratories: Springfield, VA, USA, 1977.

THANK YOU